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A comparative study of epoxy and polyurethane based coatings containing polyaniline-DBSA pigments for corrosion protection on mild steel

F.B. Diniz^{a,*}, G.F. De Andrade^a, C.R. Martins^b, W.M. De Azevedo^a

^a Departamento de Química Fundamental – CCEN – Universidade Federal de Pernambuco, 50740-560 Recife, PE, Brazil

^b Departamento de Ciências Exatas e da Terra – Universidade Federal de São Paulo, 05508-900 São Paulo, SP, Brazil

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ABSTRACT

Organic coatings based on epoxy and polyurethane matrices containing polyaniline doped with dodecyl benzene sulfonic acid (Pani-DBSA) were prepared and applied over steel plates (SAE 1020). The plates were submitted to salt spray chamber for up to 30 days in order to evaluate the corrosion protection of these coatings. The properties of the coated plates were analyzed as a function of time by electrochemical impedance spectroscopy, open circuit potential, optical microscopy, and Raman spectroscopy. In general, results indicate a decrease in the electrical resistance, increase in capacitance and decrease in open circuit potential. Epoxy based coatings have improved performance when Pani-DBSA is used as pigment, whereas for the polyurethane coatings, Pani-DBSA seems to play an adverse effect. Raman spectroscopy indicates a possible chlorination of the epoxy matrix after 30 days exposure to salt spray chamber.

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1. Introduction

The use of polyaniline as a pigment in organic coatings to prevent corrosion has been the subject of intense research [1–6]. Typical examples are composites with Fe₂O₃ and acrylic binder, epoxy plus doped and undoped polyaniline, and acrylic resin with polyaniline. Most of these coatings have been applied to steel and showed considerable improvement on corrosion protection [1,4,7–10]. Pigments for corrosion protection are sought as replacement to highly toxic heavy metal based materials. In special there is a strong drive toward the replacement of chromium based pigments. In this respect, polyaniline is a more environmentally compatible material, with a simple synthetic route [1,11–16].

The mechanism of corrosion protection proposed for these organic coatings may be classified essentially into two categories. There is an active electronic barrier model, proposed by Jain et al. [17], to explain the corrosion protection of oxide semiconductors on metals, which has been extended to polyaniline/steel interface by Sathiyarayanan et al. [3]. According to this model, an electric field at the polymer/metal interface blocks the electron transfer from the metal surface, decreasing the corrosion rate. There is also a model proposed by Wessling [18], where polyaniline acts as a catalyst for the formation of a passive metal oxide layer. The effect of polyaniline may be observed by an increase of the corrosion potential in

the positive direction. In a more recent work, Kalendová et al. [4] investigated polyaniline based coatings and concluded that electrical contact of the pigment with the metal is essential for corrosion protection in order to facilitate the several redox processes involved in the mechanism of metal oxidation/polymer reduction.

In this work, we present results of the performance of epoxy and polyurethane paints containing polyaniline pigments. The polyaniline loadings in the matrices were essayed at 0.1% and 0.5% concentration (by weight). These concentrations were chosen based on values commonly found in the literature, aiming at lower concentration levels to avoid the possibility of agglomerate formation during the step of pigment dispersion used in our procedure. Dodecyl benzene sulfonic acid (DBSA) was chosen as the dopant for polyaniline because it is known to yield a high conductivity to the polymer as well as to improve the stability and compatibility with the organic matrices used in this work [19]. The study was carried out by accelerated corrosion on saline chamber and evaluated by electrochemical impedance spectroscopy, optical microscopy and Raman spectroscopy. The effectiveness of the polyaniline coating in different matrices is evaluated.

2. Experimental methods

2.1. Polyaniline pigments synthesis and characterization

Polyaniline was chemically synthesized according with a procedure described by Haba et al. [20]. The synthesis consisted essentially on the oxidation of aniline + dodecyl benzene sulfonic

* Corresponding author. Tel.: +55 81 21268446.

E-mail address: fbd@ufpe.br (F.B. Diniz).

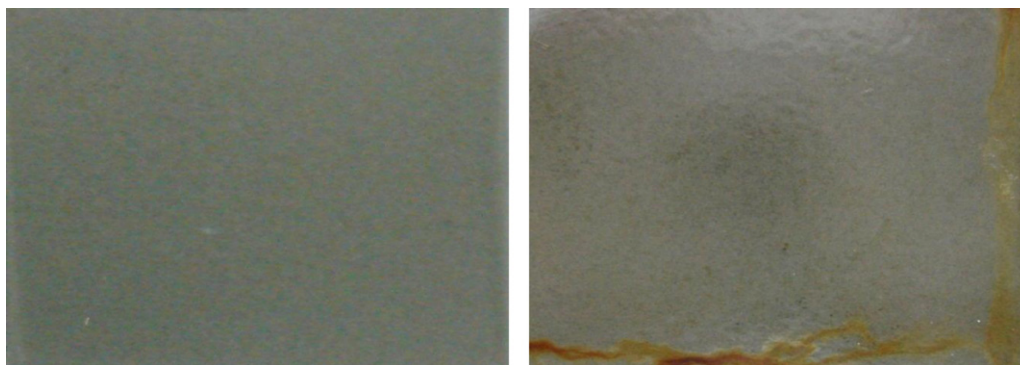


Fig. 1. Photograph of plate coated with epoxy containing 0.1% Pani-DBSA. Left, fresh plate. Right, plate after 30 days of salt spray chamber accelerated corrosion essay.

acid (DBSA) aqueous solution by ammonium persulfate at freezing temperatures. The green polyaniline powder is filtered, rinsed with acetonitrile and vacuum dried. This pigment is then classified in standard sieves (from 100 to 500 mesh). The fraction collected between 400 mesh and 500 mesh is then used as pigments in the paint formulation. The pigment was characterized by IR spectroscopy of KBr pellets.

2.2. Paint formulation

Two commercially available (epoxy and polyurethane based) paints were used as matrices for the pigments. Polyaniline was added to the resin component of the paints at two concentrations, 0.1% and 0.5%, and mechanically stirred for 30 min. Then, the hardener component (either polyamine or aliphatic isocyanate for epoxy and polyurethane, respectively) was added and followed by mechanical stirring for 20 min. Finally xylene was added and the material was stirred for 10 min and allowed to rest for 15 min. The paints were then applied to the steel plates. The epoxy painted plates were used after at least 7 days and the polyurethane painted ones, after at least 3 days. In order to evaluate the role of the pigments, paints without pigments were also prepared.

2.3. Steel plates preparation

SAE 1020 steel Plates 2 mm thick measuring 50 mm × 50 mm were jetted with glass micron spheres and cleaned with acetone under sonication, rinsed with distilled water and dried under a jet stream of air before painting. Paint was applied over these plates with a compressed air pistol aiming at a dry film thickness of 150 μm, measured with a magnetic profile meter.

2.4. Instrumental techniques

The steel plates were submitted to accelerated corrosion in a BASS model CCT-MP-GS-01/2007 cyclic salt spray chamber under the following 12 h cycle (based upon ASTM B117 standard):

Salt spray (5% NaCl) for 05 h:40 min
Drying for 00 h:20 min
Saturated moisture for 04 h:00 min
Drying for 02 h:00 min

All steps were controlled at 35 °C. For each paint formulation, a set of 6 plates were submitted to accelerated corrosion cycling.

Electrochemical impedance spectroscopy in the range of 100 mHz to 100 kHz with an amplitude of 70 mV at open circuit potential was carried out using a PAR 273A potentiostat coupled with an EG&G model 5210 lock-in amplifier. The electrochemical cell was of one compartment with Ag/AgCl, KCl sat. reference

electrode and a platinum mesh counter electrode. The working electrodes were the 25 cm² coated steel plates with working areas of 10.74 cm². All EIS measurements were carried out in 3.5% NaCl solution at room temperature.

Raman spectroscopy was carried out with a SENTERRA Brucker microscope. Samples were excited with wavelength of 785 nm from a GaAlAs diode laser with 25 mW power.

3. Results and discussion

Representative images of the plates coated with epoxy based formulations are shown in Fig. 1, for a plate coated with epoxy and 0.1% Pani-DBSA. The image on the left is for a plate before the accelerated corrosion essays, whereas on the right is shown the same plate after 30 days of accelerated corrosion in salt spray chamber. Only minor modifications can be seen, mostly due to a superficial staining of the coating. On the other hand, the Nyquist plots showed significant modifications, as seen in Fig. 2a and b, for plates coated with plain epoxy and epoxy containing 0.5% Pani-DBSA, respectively. The semi-circular shape of the curves indicates that a simple parallel RC circuit may be used, as a first approximation, to model the results. The presence of an impedance arc, rather than a straight line (characteristic of a capacitive behavior), shows the resistive component of the coating, which may be ascribed to its permeability.

A nonlinear fitting procedure was carried out using a parallel RQ circuit in series with the solution resistance. *R* corresponds to the electrical resistance of the coating, while *Q* is a constant phase element that is related to the capacitance of the coating. The lines in Fig. 2 were obtained with the parameters calculated by this fitting procedure which yielded the values of *R* and *C* shown in Table 1 for all plates investigated. With respect to the epoxy based coatings a strong decrease in resistance with time of essay, was observed (about 2 orders of magnitude), while the capacitance showed only a moderate increase (about 40%). Addition of Pani-DBSA pigment induced an increase in the resistance from $8.00 \times 10^{10} \Omega \text{cm}^2$ (for the plain epoxy coating) to $1.60 \times 10^{11} \Omega \text{cm}^2$ and $1.97 \times 10^{11} \Omega \text{cm}^2$ (for epoxy containing 0.1% and 0.5% Pani-DBSA, respectively) before the salt spray chamber experiments. The increased resistance of the epoxy coatings with Pani-DBSA persisted after 30 days of experiments. The higher resistances of the coatings with Pani-DBSA pigments indicate a better barrier protection against the salt spray environment, whereas the decrease in resistance with time may be seen as the failure of the coating as a barrier. On the other hand, the increase in capacitance is ascribed to the swelling of the coating with moisture, after exposure to the saline environment. Although the improved performances of the coatings containing Pani-DBSA pigments are quite clear (seen by the smaller capacitances), the effect of their amount, 0.1% and 0.5%, is not. There is some indication that 0.1%

Table 1

Average parameters extracted from electrochemical impedance spectroscopy for epoxy and PU coatings without and with Pani-DBSA pigments.

	Ep	Ep-0.1%	Ep-0.5%	PU	PU-0.1%	PU-0.5%
R_0 ($\Omega \text{ cm}^2$)	8.00×10^{10}	1.60×10^{11}	1.97×10^{11}	4.00×10^{10}	2.29×10^{10}	1.04×10^{10}
R_{30} ($\Omega \text{ cm}^2$)	8.43×10^8	3.95×10^9	2.44×10^9	6.79×10^9	9.81×10^8	4.05×10^6
C_0 (F cm^{-2})	6.25×10^{-11}	4.92×10^{-11}	5.03×10^{-11}	9.48×10^{-11}	1.01×10^{-10}	1.12×10^{-10}
C_{30} (F cm^{-2})	8.79×10^{-11}	7.14×10^{-11}	8.10×10^{-11}	1.01×10^{-10}	8.06×10^{-11}	1.62×10^{-10}

R_0 , initial resistance; R_{30} , resistance after 30 days in salt spray chamber. Similar notation for capacitance, C. Ep, epoxy coating; Ep-0.1%, epoxy coating with 0.1% Pani-DBSA; Ep-0.5%, epoxy coating with 0.5% Pani-DBSA; Similar notation for PU, polyurethane coatings.

Pani-DBSA pigment yields better corrosion protection, as indicated by the greater final resistance ($3.95 \times 10^9 \Omega \text{ cm}^2$), and smaller final capacitance ($7.14 \times 10^{-11} \text{ F cm}^{-2}$).

A similar discussion may be carried out for the plates coated with polyurethane matrices. Representative images of these plates are shown in Fig. 3, for a plate coated with polyurethane containing 0.1% Pani-DBSA. As in Fig. 1, fresh sample is in the left and the same sample after 30 days of accelerated corrosion essay in salt spray chamber is shown in the right. Here, the signs of corrosion are clearly seen as indicated by the formation of blisters after 30 days

Table 2

Open circuit potential for epoxy and PU coatings without and with Pani-DBSA pigments.

	Ep	Ep-0.1%	Ep-0.5%	PU	PU-0.1%	PU-0.5%
E_0 (V)	0.180	0.208	0.190	0.180	0.243	0.270
E_{30} (V)	−0.033	0.079	−0.044	0.032	−0.313	−0.410

E_0 , initial open circuit potential and E_{30} , open circuit potential after 30 days in salt spray chamber.

of assay. From Table 1, the impedance data showed 83% decrease in resistance for plain PU. For PU-0.1% Pani-DBSA and PU-0.5% Pani-DBSA the decrease in resistances were 96% and almost 100%, respectively. Therefore, the Pani-DBSA pigments clearly exerted a detrimental effect to the PU coatings. As for the capacitance results, the data did not yield a trend.

In summary, the impedance data showed that epoxy resin has improved performance with the addition of Pani-DBSA pigment. On the contrary, polyurethane showed a poorer performance upon addition of Pani-DBSA pigment. A possible explanation for these findings may be related to the aliphatic nature of the polyurethane matrix investigated. The highly polar characteristic of the aromatic system of the Pani-DBSA may cause a morphological change in the coating that leads to higher porosity of the polyurethane aliphatic matrix, which may explain the lower resistances observed. The opposite may be said of the epoxy system. The aromatic nature of the pigment was compatible with the epoxy matrix leading to good interaction and improving the barrier effect of the epoxy network.

At the start of each impedance measurement, open circuit potentials (OCP) were measured. Their values are shown in Table 2. In all cases investigated, there is a decrease in open circuit potential. This behavior can be interpreted, in terms of an increase in corrosion rate with time in the saline environment. The protective effect of the coating decreases, in accordance with the analysis presented above. The initial values of OCP (E_0) were higher for coatings containing Pani-DBSA, the higher values being found in the PU coatings. After 30 days in the salt spray chamber the open circuit potentials decreased the most for the PU coatings that contained Pani-DBSA pigments. The highest open circuit potential was displayed for the epoxy coating with 0.1% Pani-DBSA pigment. Therefore, although the PU initial result was better (in terms of open circuit potential), it degraded significantly to the lowest value found in all cases. These results are analogous to those found in the impedance measurements. Again, it seems that the Pani-DBSA adversely affects the performance of the epoxy based coating. Azim [21] and Sathiyarayanan [22] have measured open circuit potentials along with corrosion measurements and found an initial decay in OCP values. However, these authors found a posterior increase in OCP. They assigned this increase in values to the formation of a passive layer on iron surface by Pani. These authors have used a larger amount of Pani (>1%) than presented here. Therefore, it is reasonable to postulate that with the lower Pani loadings, there is no passive layer formation, as evidenced by the decrease in OCP values with time. For the present work, results may be better interpreted in terms of the model proposed by Jain et al. [17], of an active electronic barrier created by the

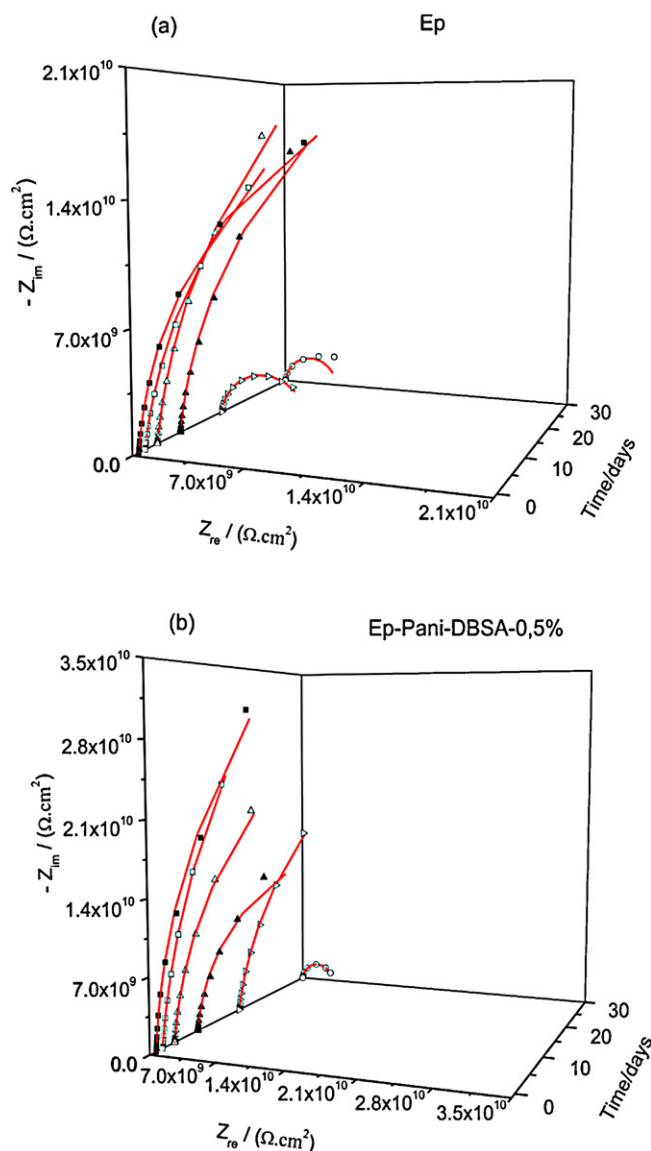


Fig. 2. Nyquist plots representative for the epoxy coating (a) and modified epoxy coating with 0.5% Pani-DBSA (b) for 0, 1, 3, 7, 15 and 30 days of exposure to salt spray chamber. Solid lines correspond to calculated values.

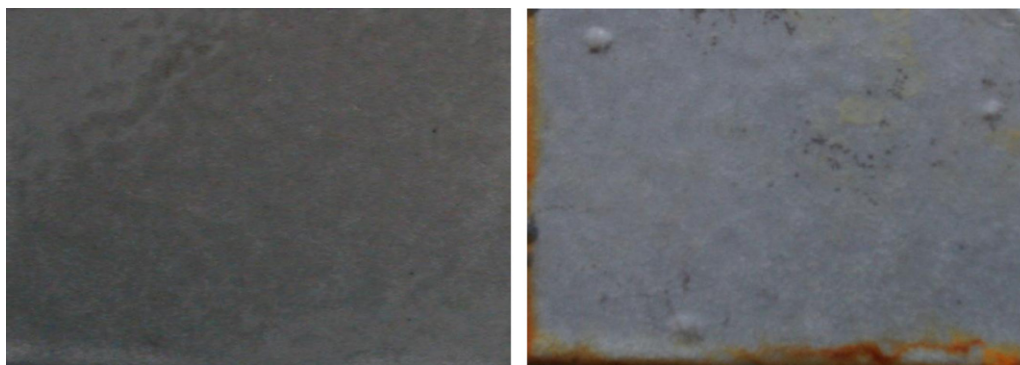


Fig. 3. Photograph of plate coated with polyurethane containing 0.1% Pani-DBSA. Left, fresh plate. Right, plate after 30 days of salt spray chamber accelerated corrosion essay.

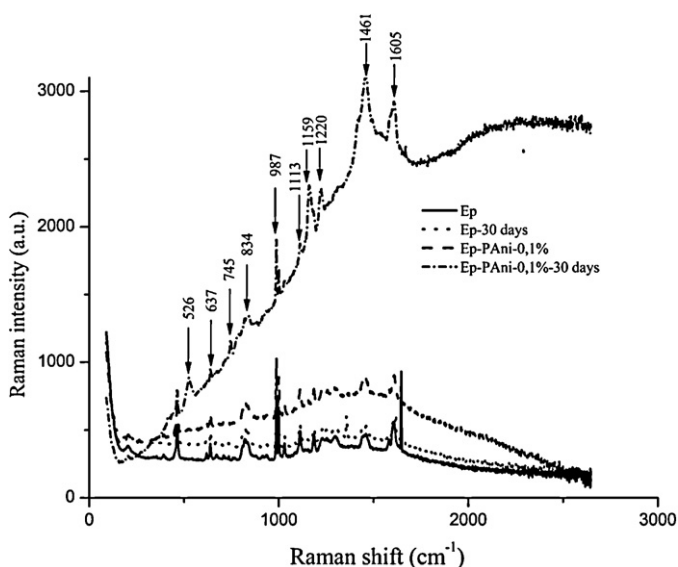


Fig. 4. Raman spectra for epoxy and epoxy containing 0.1% Pani-DBSA coatings for fresh samples and samples exposed for 30 days in salt spray chamber.

Pani present in the coating as a pigment. In the case of polyurethane coatings, this barrier seems to fail with increasing Pani loading, leading to failure.

Raman Spectroscopy results for epoxy base coatings are shown in Fig. 4 and the assignments are given in Table 3. There are some noticeable changes comparing the fresh material with those

Table 3
Raman bands assignments for epoxy coating.

Band position (cm ⁻¹)	Assignments	References
1605	C–C stretching vibration of aromatic; C–O stretching vibration of amide; skeletal vibrations of C=C double bonds in aromatic ring	[23,24]
1461	Stretching vibration p-substituted benzene	[23]
1220	C–O stretching vibration ether bridge	[23]
1159	C–O stretching vibration of secondary alcohol	[23]
1113	Aromatic C–H stretching and in plane deformation	[25]
987	Epoxy group	[25]
745	C ₃ –C ₂ skeletal	[25]
834	Out of plane bending of aromatic C–H	[23,24]
637	Aromatic ring vibrations (p-substituted benzene); aromatic C–H out of plane deformation	[25,26]
526	C–Cl stretching	[27,28]

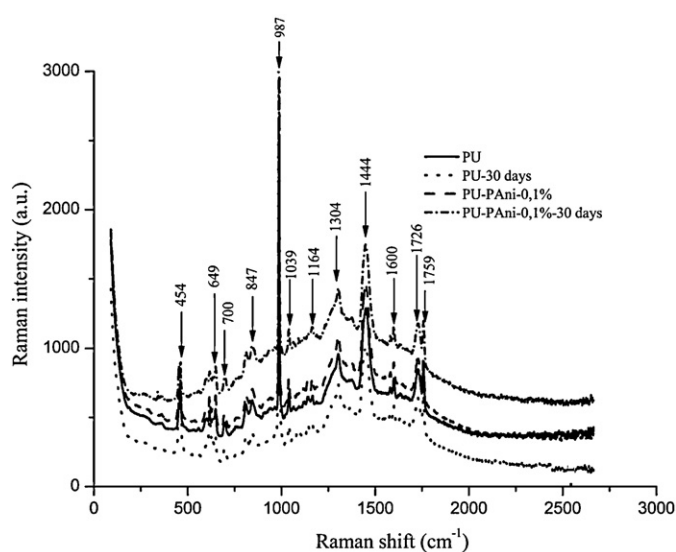


Fig. 5. Raman spectra for polyurethane and polyurethane containing 0.1% Pani-DBSA coatings for fresh samples and samples exposed for 30 days in salt spray chamber.

after 30 days in the salt chamber, specially the increase in the background signal (much more pronounced in the coatings containing 0.1% Pani-DBSA), due to increased fluorescence. Moreover, in epoxy coating containing 0.1% Pani-DBSA, there are some additional peaks, specially, one at 526 cm⁻¹. These results indicate that some structural changes, associated with the pigment, occurred in the polymer. The presence of the 526 cm⁻¹ is tentatively assigned to C–Cl stretching [27,28], which may be ascribed to chlorination of the epoxy matrix. The absence of such modifications in the epoxy sample without pigment may be an indication that the structural changes observed in the epoxy matrix are induced by the pigment and are such that they improve the corrosion resistance of the polymer. For the polyurethane matrix there are no noticeable changes in the Raman spectra (Fig. 5). Together with the previous results, Raman spectroscopy indicates some chemical interaction of the epoxy resin with the polyaniline pigment. This interaction may be the reason for the better performance of the 0.1% Pani-DBSA epoxy coating. In the case of polyurethane coatings, there is no clear interaction and the pigment may disrupt the polyurethane matrix leading to a worsening on corrosion protection.

4. Conclusion

The results presented above showed that Pani-DBSA pigments increase the performance of epoxy based coatings regarding corrosion resistance under salt spray chamber essays. Epoxy with

Pani-DBSA displayed higher resistances pointing to a better performance as barrier for salt spray environment. Open circuit potentials also pointed in the same direction. Since OCP values decreased with time to salt spray chamber exposure, there seems to be no passivation effect induced by the coating, and the active electronic barrier model seems to be a more reasonable explanation for the mechanism of protection. Polyurethane coatings without Pani-DBSA pigment yielded good corrosion protection. However, addition of Pani-DBSA pigments to this matrix at 0.5% level degraded the coating such that its resistance, measured by electrochemical impedance spectroscopy, decreases about 4 orders of magnitude (from $1.04 \times 10^{10} \Omega \text{ cm}^{-2}$ to $4.05 \times 10^6 \Omega \text{ cm}^{-2}$). Therefore, it was possible to postulate that the interactions of Pani-DBSA pigments with the matrices are different according to the matrix used. Raman spectroscopy gives some evidence of chlorination of the epoxy matrix containing 0.1% Pani-DBSA, caused by exposure to chloride ions in the salt spray chamber.

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